

Complex Numbers: Cartesian Form & Algebra



Basic arithmetic with complex numbers

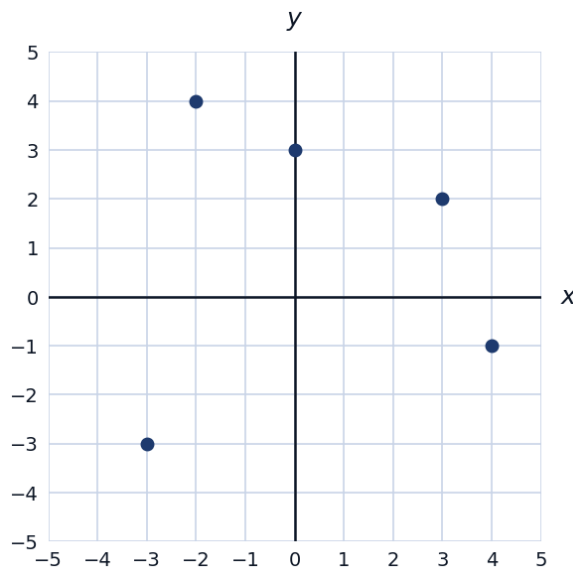
- 1 Given $z_1 = 3 + 2i$ and $z_2 = 1 - 4i$, find $z_1 + z_2$ and $z_1 - z_2$.
 - 2 Find $z_1 z_2$ for $z_1 = 3 + 2i$ and $z_2 = 1 - 4i$.
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The complex conjugate and division

- 3 Find the complex conjugate of $z = 5 - 3i$, and compute $z\bar{z}$.
 - 4 Simplify $\frac{2+i}{3-2i}$ by multiplying by the conjugate of the denominator.
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Modulus and the Argand diagram

- 5 Find the modulus $|z|$ of $z = 6 - 8i$.
- 6 The graph shows several complex numbers plotted as points on the Argand diagram (real part on the horizontal axis, imaginary part on the vertical axis).



- a Read off the complex number plotted closest to the point (3, 2) in Cartesian (real, imaginary) form.
 - b Find the modulus of the complex number plotted at $(-3, -3)$.
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Solving equations with complex roots

- 7 Solve $x^2 + 4x + 13 = 0$, giving your answer in the form $a + bi$.
- 8 One root of a quadratic equation with real coefficients is $2 - 5i$. State the other root, and find the quadratic equation.
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Equating real and imaginary parts

- 9 Find real numbers x and y such that $(x + yi)(2 - i) = 8 + i$.
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Powers of complex numbers using algebra

- 10 Find z^2 and z^3 for $z = 1 + i$, using direct algebraic expansion (without polar form).
- 11 Simplify i^{23} , using the fact that powers of i cycle with period 4 ($i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$).
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The geometry of complex conjugates and reflections

- 12 Explain, in terms of the Argand diagram, what geometric transformation taking the conjugate of a complex number represents, and state the conjugate of $z = -4 + 7i$.
- 13 If $z = a + bi$, express $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$ in terms of z and $\bar{z}$.
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Solving simultaneous equations with complex numbers

- 14 Solve the simultaneous equations $z_1 + z_2 = 5 + 3i$ and $z_1 - z_2 = 1 + 7i$ for complex numbers z_1 and z_2 .
- 15 Given that $z = 2 + 3i$ is a root of $z^3 - 5z^2 + 17z - 13 = 0$ (a cubic with real coefficients), state the second complex root immediately (without further calculation), then find the third (real) root.
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Loci and regions in the complex plane

- 16 Describe, geometrically, the locus of points z satisfying $|z - 3 - 4i| = 5$, and state its center and radius.
- 17 Find the two square roots of $z = 5 + 12i$ algebraically, by writing $\sqrt{5 + 12i} = a + bi$ and solving the resulting simultaneous equations $a^2 - b^2 = 5$ and $2ab = 12$.

Complex number identities and proof

- 18** Prove that $|z_1 z_2| = |z_1| |z_2|$ for any complex numbers $z_1 = a + bi$ and $z_2 = c + di$, by expanding both sides directly.
- 19** Given $z = x + yi$ satisfies $z^2 = \bar{z}$, find all complex solutions z .